

EXPERIMENT - 3 (16-BIT ADDITION)

Aim: To write an assembly language program to implement 16-bit addition using 8086 processor.

Algorithm:

1. Start the program by loading a register pair with address of 1st number.
2. Copy the data to another register pair.
3. Load the second number to the first register pair.
4. Check for carry.
5. Add the two register pair contents.
6. Store the value of sum and carry in memory locations.
7. Terminate the program.

Program:

MNEMONICS	Explanation.
LDA 3050	Load the content of the memory location 3050H into the accumulator (A).
MOV B, A	Move the content of the accumulator (A) into register B. Now, register B contains the value.
LDA 3051	Load the content of the memory location 3051H into the accumulator (A).
ADD B	Add the content of register B (value from 3050H) to the accumulator (A).
STA 3052	Store the result of the addition (from the accumulator) into memory location.
LDA 3053	Load the content of the memory location 3053H into the accumulator (A).
MOV B, A	Move the content of the accumulator (A) into register B.
LDA 3054	Load the content of the memory location 3054H into the accumulator (A).
ADD B	Add the content of register B to the Accumulator (A).
STA 3055	Store the result of the addition into memory location 3055.
HLT	Halt the execution of the program.

Input:

Address	Data
---------	------

3050	2
------	---

3051	3
------	---

3053	5
------	---

3054	5
------	---

Output:

Address	Data
---------	------

3052	5
------	---

3055	10
------	----

0184

[Signature]

Result: Thus the program was executed successfully using 8085 processor simulator.

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

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Registers

A	0A	
BC	05	00
DE	00	00
HL	00	00
PSW	00	00
PC	42	17
SP	FF	FF
Int-Reg	00	

Flag

S	0
Z	0
AC	0
P	1
C	0

Load me at

1	LDA	3050
2	MOV	B,A
3	LDA	3051
4	ADD	B
5	STA	3052
6	LDA	3053
7	MOV	B,A
8	LDA	3054
9	ADC	B
10	STA	3055
11	HLT	

Decimal - Hex Conversion

Decimal

Hex

0

0

To Hex

To Dec

I/O Ports

0

-

+

00

Update Port Value

Memory

3054

-

+

05

Update Memory

DataStackKeypadMemoryI/O Ports

Start3052OK

Address (Hex)	Address	Data
0BEC	3052	5
0BED	3053	5
0BEE	3054	5
0BEF	3055	10
0BF0	3056	0
0BF1	3057	0
0BF2	3058	0
0BF3	3059	0
0BF4	3060	0
0BF5	3061	0
0BF6	3062	0
0BF7	3063	0
0BF8	3064	0
0BF9	3065	0
0BFA	3066	0
0BFB	3067	0

Line NoAssembler Message

0Program assembled successfully